ABC Call Volume Trend Analysis

Final Project – 4

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Project Description: -

In this project, you'll be diving into the world of Customer Experience (CX) analytics, specifically focusing on the inbound calling team of a company. You'll be provided with a dataset that spans 23 days and includes various details such as the agent's name and ID, the queue time (how long a customer had to wait before connecting with an agent), the time of the call, the duration of the call, and the call status (whether it was abandoned, answered, or transferred).

A Customer Experience (CX) team plays a crucial role in a company. They analyze customer feedback and data, derive insights from it, and share these insights with the rest of the organization. This team is responsible for a wide range of tasks, including managing customer experience programs, handling internal communications, mapping customer journeys, and managing customer data, among others.

In the current era, several AI-powered tools are being used to enhance customer experience. These include Interactive Voice Response (IVR), Robotic Process Automation (RPA), Predictive Analytics, and Intelligent Routing.

One of the key roles in a CX team is that of the customer service representative, also known as a call center agent. These agents handle various types of support, including email, inbound, outbound, and social media support.

Inbound customer support, which is the focus of this project, involves handling incoming calls from existing or prospective customers. The goal is to attract, engage, and delight customers, turning them into loyal advocates for the business.

File contains -

No. of Rows – 117989

Columns – 13

Blank cells – 47877

No. of #N/A in Agent_Name & Agent_ID Column – 34198

Tools used: -

MS Excel (Office365)

MS Word (Office365)

Data Analytics Tasks: -

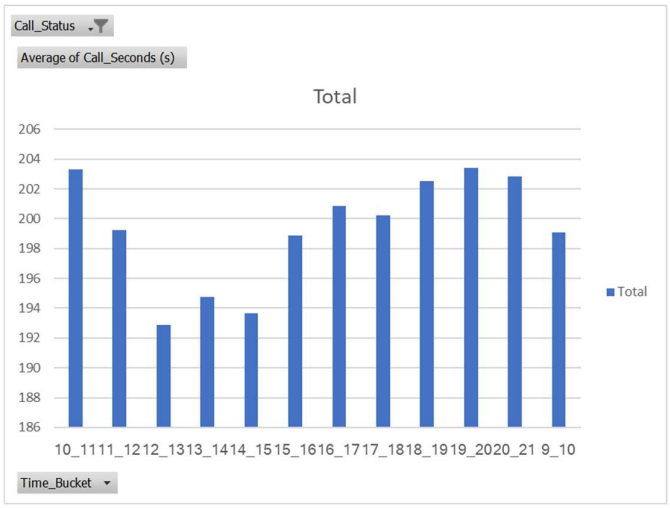
You have been provided with a dataset that contains information about the inbound calls received by a company named ABC, which operates in the insurance sector. Your task is to use this data to answer the following questions: -

1. **Average Call Duration:** Determine the average duration of all incoming calls received by agents. This should be calculated for each time bucket.

Your Task: What is the average duration of calls for each time bucket?

Call_Status	answered

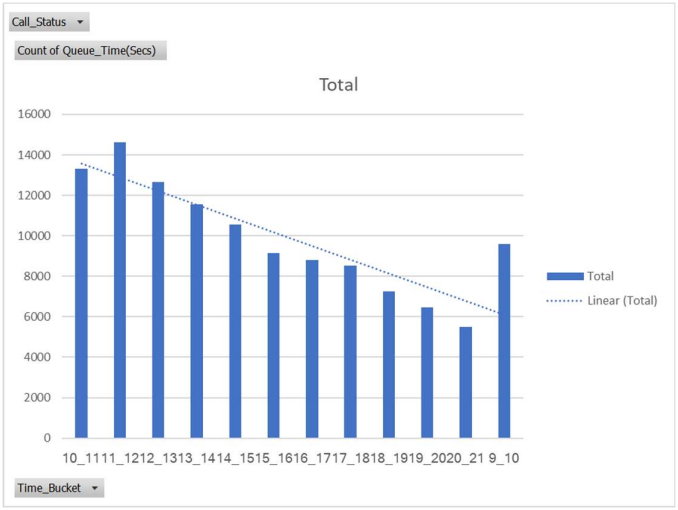
Time_Bucket	Average of Call_Seconds (s)
10_11	203.33
11_12	199.26
12_13	192.89
13_14	194.74
14_15	193.68
15_16	198.89
16_17	200.87
17_18	200.25
18_19	202.55
19_20	203.41
20_21	202.85
9_10	199.07
Grand Total	**198.62**



2. **Call Volume Analysis:** Visualize the total number of calls received. This should be represented as a graph or chart showing the number of calls against time. Time should be represented in buckets (e.g., 1-2, 2-3, etc.).

Your Task: Can you create a chart or graph that shows the number of calls received in each time bucket?

Row Labels	Count of Queue_Time(Secs)
10_11	13313
11_12	14626
12_13	12652
13_14	11561
14_15	10561
15_16	9159
16_17	8788
17_18	8534
18_19	7238
19_20	6463
20_21	5505
9_10	9588
Grand Total	**117988**



3. **Manpower Planning:** The current rate of abandoned calls is approximately 30%. Propose a plan for manpower allocation during each time bucket (from 9 am to 9 pm) to reduce the abandon rate to 10%. In other words, you need to calculate the minimum number of agents required in each time bucket to ensure that at least 90 out of 100 calls are answered.

Your Task: What is the minimum number of agents required in each time bucket to reduce the abandon rate to 10%?

Action Taken: -

For this first I created pivot table using date & Time in row and count of call duration in values.

Then Calculated average and percentage as per call status using the pivot table data.

Call Status	Average	percentage
Abandon	1496	29%
Answer	3585	70%
Transfer	49	1%
Grand Tota	**5130**	**100%**

Sr. No.	Assumption
1	An agent works for 6 days a week;
2	On average, each agent takes 4 unplanned leaves per month;
3	An agent's total working hours are 9 hours, out of which 1.5 hours are spent on lunch and snacks in the office.
4	On average, an agent spends 60% of their total actual working hours (i.e., 60% of 7.5 hours) on calls with customers/users.
5	The total number of days in a month is 30.

Then as per given assumption calculated using formulas: -

- **Working hours per agent: -** (=(60/100)*7.5) **=** 4.5
- **Average call duration: -** 198.62 (calculated in Task 2 using pivot Table)
- **Total no of working hours needed to reach 90% : -** (Total average*Average call duration*0.9)/3600

  (5130*198.2*0.9)/3600 = 254.73
- **No. of agents required (9 AM – 9 PM): -** (Total no of working hours needed to reach 90%)/working hours per agent

  I.e 254/4.5 = 57

1	working hours per agent	4.5
2	Average call duration	198.62
3	Total no of working hours needed(90%)	254.73
4	No. of agents needed(9AM-9PM)	57

4. **Night Shift Manpower Planning**: Customers also call ABC Insurance Company at night but don't get an answer because there are no agents available. This creates a poor customer experience. Assume that for every 100 calls that customers make between 9 am and 9 pm, they also make 30 calls at night between 9 pm and 9 am. The distribution of these 30 calls is as follows:

Your Task: Propose a manpower plan for each time bucket throughout the day, keeping the maximum abandon rate at 10%.

Action Taken: -

For this first I created pivot table using date & Time in row and count of call duration in values.

Then Calculated average and percentage as per call status using the pivot table data.

Call Status	Average	percentage
Abandon	1496	29%
Answer	3585	70%
Transfer	49	1%
Grand Tota	**5130**	**100%**

Sr. No.	Assumption
1	An agent works for 6 days a week;
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4	On average, an agent spends 60% of their total actual working hours (i.e., 60% of 7.5 hours) on calls with customers/users.
5	The total number of days in a month is 30.

Then as per given assumption calculated using formulas: -

- **Working hours per agent: -** (=(60/100)*7.5) **=** 4.5
- **Average call duration: -** 198.62 (calculated in Task 2 using pivot Table)
- **Average no of calls in night shift: -** (Total average*0.3)

  5130*0.3 = 1539
- **Total no of working hours needed to reach 90% in Night shift : -**

  (Average no of calls in night shift *Average call duration*0.9)/3600

  (1539*198.2*0.9)/3600 = 76
- **No. of agents required (9 PM – 9 AM): -**

  (Total no of working hours needed to reach 90% in night shift)/working hours per agent

  I.e 76/4.5 = 17

1	working hours per agent	4.5
2	Average call duration	198.62
3	Average no of calls in Night shift	1539
4	Total no of working hours needed(90% in night shift)	76
5	No. of agents needed in Night shift(9PM-9AM)	17

File Uploaded link –

https://docs.google.com/spreadsheets/d/1ORiGTRZUh6sFiKxcdsZ_WIeCCwBzUud7/edit?usp=drive_link&ouid=107355046625792303579&rtpof=true&sd=true

~~END~~